

Research Paper

Confronting the Amazon's fire crisis: Evidence of early fire occurrence and increased atmospheric gases from the GOSAT satellite

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ABSTRACT

Forest fires represent a significant threat to the Amazon biome, contributing to elevated greenhouse gases (GHGs) emissions. This study analyzes temporal trends in column-averaged concentrations of CO₂ (XCO₂) and CH₄ (XCH₄) in the atmosphere using satellite observations from GOSAT, while exploring the relationships between surface spectral indices and wildfire activity in the region. Key variables included the Enhanced Vegetation Index (EVI), Fire Radiative Power (FRP), and MODIS-based fire foci data. Additionally, Sun-Induced Fluorescence (SIF), XCH₄, and XCO₂ data were obtained from GOSAT for the period 2009–2019, alongside global forest fire emissions from the VIIRS-based Fire Emission Inventory (VFEI) for 2012–2019. Our findings indicate significant increases in monthly averages of XCO₂ (>1.3 ppm) starting in April 2013, identified as a change point using the Pettitt test. Similarly, FRP values increased for April (>12 MW) and July (>23 MW), suggesting an earlier onset of wildfire activity and subsequent atmospheric impacts. These results align with VFEI emissions data, which also show rising GHG levels during the study period. The observed increases in wildfires and associated XCO₂ concentrations are likely linked to anthropogenic activities, particularly land-use changes, underscoring the critical role of human influence in exacerbating GHGs emissions within the Amazon biome. These findings highlight the urgent need for sustainable land management strategies to mitigate the adverse impacts of wildfires and preserve the biome's role as a vital carbon sink.

1. Introduction

Tropical forest fires are significant contributors to greenhouse gas (GHG) emissions, profoundly impacting terrestrial ecosystems on both regional and global scales (Miettinen et al., 2016; Ribeiro et al., 2020). These fires disrupt biogeochemical cycles, alter atmospheric composition, and degrade the functional attributes of terrestrial ecosystems, amplifying their negative effects on the global climate system (Koch et al., 2019; Lovejoy & Nobre, 2019; Williams et al., 2016). The suppression of forests, driven by deforestation and land-use changes, is recognized as a critical factor in global environmental transformations (Koch et al., 2019).

The prevalence of large-scale fires in tropical forests, such as the Amazon, has become a pressing environmental concern. In 2020 alone,

Brazil recorded 222,797 fire foci, with the Amazon accounting for 46.3 % (103,161 foci) of these incidents (INPE, 2022). This upward trend in fire occurrences within the Amazon biome (Silva et al., 2021a) is closely associated with a combination of factors, including a warmer climate, more severe and frequent droughts, and continuous land-use changes in the region (Oliveira Filho et al., 2022). The majority of these fires are anthropogenic, often initiated by farmers clearing deforested areas for cattle ranching. The lack of effective strategies to combat these fires exacerbates their impacts, particularly in biomes and protected areas across the region (Oliveira Filho et al., 2022).

Under the Paris Climate Agreement, Brazil has committed to restoring 120,000 km² of forests by 2030 (Lovejoy & Nobre, 2019). However, recent trends reveal a concerning increase in deforestation, with deforested areas in the Legal Amazon rising by 9.5 %, from 10,129

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km² in 2019 to 11,008 km² in 2020 (INPE, 2022). Alarming, deforestation in the Brazilian Amazon contributes approximately 200 t ha⁻¹ of CO₂ emissions (Silva et al., 2020; Soares Neto et al., 2009).

While various methods exist to measure GHG emissions—such as ships, balloons, towers, and aircraft—these approaches often present spatial and temporal limitations due to their high costs (Gloor et al., 2000). Consequently, uncertainties persist in estimating CO₂ sources and sinks (Mousavi et al., 2017; Yoshida et al., 2011). Advances in satellite-based remote sensing have proven invaluable in addressing these gaps, providing comprehensive data on GHG distributions and fluxes (Wecht et al., 2014).

The Japanese Space Agency (JAXA) GOSAT mission (Kuze et al., 2009) and NASA's OCO-2 mission (Frankenberg et al., 2015), along with other satellite initiatives, enable the measurement of column-averaged dry air mole fractions of CO₂ and CH₄ (XCO₂ and XCH₄) using advanced sensors such as the TIR/SWIR Fourier Transform Spectrometer (TANSO-FTS) and the Short-Wave Infrared (SWIR) bands (Kuze et al., 2009). Remote sensing applications have demonstrated efficacy in estimating CO₂ emissions from fires across diverse ecosystems, including boreal forests (Guo et al., 2017), tropical forests (Heymann et al., 2017; Jiang et al., 2021; Liu et al., 2017a; Yin et al., 2018), arid regions (Li et al., 2019). Additionally, research enables a better understanding of the sources and sinks of CH₄ and CO₂ (Mousavi et al., 2024a, 2024b).

Previous studies on Amazonian fires and atmospheric gases have been limited by (i) coarse temporal resolution focusing mainly on annual patterns, (ii) insufficient integration of FRP, SIF, and atmospheric GHG data, and (iii) a lack of detailed assessment of early dry-season fire dynamics. These limitations hinder our understanding of how fire seasonality is shifting and its implications for the regional carbon balance.

This study provides the first integrated multi-sensor analysis of early fire occurrence (April and July) in the Brazilian Amazon, linking FRP trends from MODIS and VIIRS with atmospheric XCO₂ anomalies detected by GOSAT over an 11-year period. Unlike previous research, which has mainly focused on peak dry-season fires, our work reveals a significant shift towards earlier fire onset, with potential implications for carbon fluxes, biome resilience, and policy design for fire management.

Therefore, the present study aims to answer the following questions: (i) Is there a relationship between the GHG orbital data and SIF (vegetation) and FRP (fire power)? (ii) What are the temporal trends in the concentration of CO₂ in the Amazon biome during the study period?

2. Materials and methods

2.1. Study area

The study area comprises the Brazilian Amazon Forest biome. Territorial representation exceeds 4.2 million km², covering nine Brazilian states: Acre, Amapá, Amazonas, Pará, Roraima, Rondônia and partially the state of Maranhão, Mato Grosso and Tocantins (IBGE - Instituto Brasileiro de Geografia e Estatística, 2020) (Fig. 1).

The Brazilian Amazon Forest is a critical component of global ecological balance, offering essential ecosystem services such as unparalleled biodiversity, climate regulation at regional and global scales, and the preservation of one of Earth's largest freshwater reserves (Strand et al., 2018). Despite its significance, the Amazon has faced alarming rates of deforestation over the past four decades. Between 1975 and 2018, approximately 20 % of the forest's total area—equivalent to 788,353 km²—was deforested (da Cruz et al., 2021).



Fig. 1. Location map of the Brazilian Amazon Forest biome.

Much of this deforestation is attributed to land conversion for livestock production, with the state of Pará being the most affected. In Pará, 35.2 % of the deforested land was converted into pasture, marking it as the most severely impacted state within the Brazilian Amazon (Souza et al., 2020). These figures underscore the critical need for sustainable land-use policies to mitigate ongoing deforestation and its environmental consequences in this vital ecosystem.

The study area encompasses the Brazilian Amazon biome, characterized by diverse climatic conditions as classified under the Köppen Climate Classification system. The region predominantly features three climate types: Af (tropical rainforest, no dry season), Am (tropical monsoon), and Aw (tropical savanna with dry winters). Annual rainfall across the biome averages over 1,900 mm, with some areas receiving up to 4,000 mm, and the mean annual temperature is approximately 26.8 °C (Alvares et al., 2013). The hydrological cycle of the region is marked by distinct seasonal patterns, with the wet or rainy season spanning from October to April and the dry season extending from May to September (Zemp et al., 2017).

This study focuses on understanding the environmental dynamics of the Brazilian Amazon through an integrative approach, linking climatic variability, vegetation patterns, and fire occurrences. The climatic context provided by these parameters informs the analysis of GHGs emissions and deforestation trends, with a particular focus on their implications for regional and global ecological stability.

2.2. Data acquisition of fire radiative power and fire foci

Active fire detection and fire radiative power (FRP) measurements were employed to determine the timing, location, and radiative intensity of fires occurring in the Amazon Forest. FRP serves as a robust proxy for estimating biomass burning, a concept initially introduced by Kaufman et al. (1998) and subsequently refined by Wooster et al. (2003). FRP represents the total radiative energy emitted by a fire, as governed by the Stefan-Boltzmann law, and is typically expressed in megawatts (MW). For this study, the FRP of fire-affected pixels was calculated using an algorithm designed to identify areas with high infrared radiation emissions indicative of active combustion (Giglio et al., 2003).

To detect and characterize fire activity, we utilized data products (MYD14 and MOD14) from the Moderate Resolution Imaging Spectroradiometer (MODIS) sensor aboard the Terra and Aqua satellites, as well as data from the Visible Infrared Imaging Radiometer Suite (VIIRS) on the Suomi National Polar-orbiting Partnership (Suomi NPP) and NOAA-20 satellites (Schroeder et al., 2022). Fire event data were obtained from the Fire Information for Resource Management System (FIRMS), accessible via NASA's EarthData portal (<https://earthdata.nasa.gov/firms>; accessed October 2, 2021). FIRMS provides near-real-time fire detection data with a temporal resolution of 3 h and a spatial resolution of 1 km. For this analysis, fire detections were aggregated into monthly averages spanning April 1, 2009, to December 31, 2019.

To evaluate the emission of trace gases from global forest fires, we employed the VIIRS-based Fire Emission Inventory (VFEI) dataset for the period 2012–2019 (Ferrada et al., 2022). This Inventory (VFEI) is constituted by the emissions of trace gases by applying the top-down approach. This inventory uses a bottom up approach to estimate emissions of trace gases, including CO₂, CH₄, and carbon monoxide, by combining FRP measurements with emission factors (EFs). The FRP estimates were derived from the VIIRS I-band sensor data (Schroeder et al., 2022), while the EFs were obtained from Andreae (2019). The VFEI provides daily emission fluxes for various trace gases and aerosols with a spatial resolution of 0.005°. The emissions were calculated using the following equation:

$$E_s = k EF_s \frac{FRP}{A} \quad (1)$$

Where: E_s emissions for species s (trace gases) in $\text{kg m}^{-2} \text{s}^{-1}$; k is a constant between the amount of dry biomass burnt and FRP (kg of dry matter per MJ); EF_s is the emission factor for a given trace gases (grams trace of gases per kg dry matter); $\frac{FRP}{A}$ the ratio between FRP (MW; or MJ s^{-1}) and the satellite pixel area or A (m^2).

2.3. Acquisition of Enhanced Vegetation Index data

The Enhanced Vegetation Index (EVI) was proposed by Huete et al. (1997), which minimises the effect of the ground and atmosphere considering the blue spectral band. The images used were retrieved from the MOD13C2 product of collection 6 from the MODIS sensor, with an aggregation of 16 days and 1 km of spatial resolution, where the EVI values, estimated by Equation (2) (Huete et al., 2002). The retrieval and processing of the data were performed in Google Earth Engine for the time series of the study. The images obtained were multiplied by their scale factor of 0.0001, and thus, their values ranged from -1 to 1 .

$$EVI = g \frac{\rho_{IVP} - \rho_V}{\rho_{IVP} + (c1 \cdot \rho_V) - (c2 \cdot \rho_A) + L} \quad (2)$$

Where ρ_{IVP} , ρ_V , and ρ_A are the reflectance in the near-infrared spectral range, red and blue, respectively. g is the gain factor (2.5), $c1$ and $c2$ are the correction coefficients for the atmospheric effects of red (6) and blue (7.5), respectively, and L is the factor of the correction for soil interference (1).

2.4. Rainfall data

The Climate Hazards Group Infrared Precipitation with Stations (CHIRPS) data were obtained through the Google Earth Engine platform using JavaScript codes for image acquisitions '.TIFF' format, with a spatial resolution of 0.05° for the dataset since 1981 from the collection ("UCSB-CHG/CHIRPS/DAILY") (Funk et al., 2015). In the time series of the images, an annual sum was made in each pixel for each year of the study.

2.5. Acquisition of XCO₂ and XCH₄ data from the GOSAT satellite

On January 23, 2009, the GOSAT satellite was launched in Japan. It is important to highlight the two main instruments on board: 1) a thermal sensor (TIR) and a near-infrared sensor (SWIR) for GHG monitoring (TANSO), with the Fourier Transform Spectrometer technique (FTS), 2) Cloud and Aerosol Imager (CAI), respectively. The TANSO-FTS sensor has four bands: three SWIR bands (in the spectral range of 760, 1600, and 2000 nm) used in estimating the average CO₂ column concentrations (XCO₂) and a TIR band (5500–14300 nm) used to describe the vertical profile of CO₂ concentration in the upper troposphere. The images from the TANSO-CAI sensors have four bands (0.38, 0.68, 0.87, and 1.6 μm) providing a spatial resolution of 500 m, with application primarily for mapping the properties of clouds and aerosols in the atmosphere (Guerlet et al., 2013; Ross et al., 2013).

The main objective of the GOSAT mission was to assess the sources and sinks of GHGs, mainly to estimate the concentrations of carbon dioxide and methane (XCO₂ and XCH₄, in ppm), on a regional and global scale. In this way, it contributes to environmental managers and researchers on the understanding of the carbon cycle of ecosystems (Guo et al., 2012, 2017; Oguma et al., 2011; Taylor et al., 2022).

The XCO₂ data were taken from Earthdata (NASA) from the site <https://disc.gsfc.nasa.gov/datasets/ACOS_L2S_9r/summary> with files in netcdf format (Taylor et al., 2022).

The column-averaged dry air mole fractions of methane data (XCH₄) were obtained from the University of Leicester GOSAT Proxy, version 9.0, and it is important to note that they are in the public domain. (https://data.ceda.ac.uk/neodc/gosat/data/ch4/nceov1.0/CH4_GOS_OCP/), in netcdf format (Parker, R.; Boesch, 2020). The XCH₄ retrieval

uses the CO₂ proxy method (R. Parker et al., 2011; R. J. Parker et al., 2015). This is due to the XCO₂ fluctuates or vary much less than the XCH₄. Furthermore, the spectral proximity of the absorption band of CO₂ with CH₄, can be used as a proxy method to minimizing scattering effect of aerosols. For this, they used the rate of the column-averaged dry air mole fractions of CO₂ and CH₄ and multiply by CO₂ global chemistry transport model: GEOS-Chem (Feng et al., 2011), Carbon Tracker (Peters et al., 2007) or LMDZ (Chevallier et al., 2010), according with equation (3):

$$\text{Proxy XCH}_4 : \frac{[\text{XCH}_4]}{[\text{XCO}_2]} * \text{ModelXCO}_2 \quad (3)$$

This GOSAT XCH₄ Proxy retrieval approach has various advantages over the full physics approach and has been validated with aircraft measurements in the Amazon region (Tunnicliffe et al., 2020; Wilson et al., 2020).

2.6. Sun-induced chlorophyll fluorescence (SIF757) from GOSAT

The sun-induced chlorophyll fluorescence (SIF) is the emission of photons by the chlorophyll molecules during the photosynthetic process of the plant, and in the two photosystems (I and II), and the emission ranges are from 600 to 850 nm (Castro et al., 2020). The SIF data enable to track photosynthetic activity in the ecosystem in regional and global scale (Drusch et al., 2017). Phenological seasonality in Amazonian ecosystems is conditioned by climatic drivers. Thus, the seasonality of the SIF in the Amazon biome is answered with 76 % of the seasonal variation of radiation, on the other hand 13 % and 11 % are answered by rainfall and a combination of both (Bertani et al., 2017).

The first satellite observations of global sun-induced chlorophyll fluorescence (SIF) came from GOSAT (Frankenberg et al., 2011; Joiner et al., 2011). Thus, data from SIF of 757 nm level 2 with GOSAT Lite version 9 (v9) (Doughty et al., 2022) were obtained in netCDF format files available at <https://data.caltech.edu/records/8771> (Last accessed: April 22, 2022).

Bioclimatic variables were derived from the monthly maximum temperature and wind speed provided by WorldClim data sets (Hijmans et al., 2005). Summary information for the different variables is used to identify the tendency of GHGs and the fire foci, their source and their temporal and spatial resolution. Thus, the description of the dataset to variables was described in Table 1.

2.7. Pre-process of the data

To understand the natural and regional variability of XCO₂ and XCH₄ and their relationship with biophysical variables, a detrending regression method was employed (Gujarati, 2009). The detrending process is an important step to reduce background and atmospheric transport influence (da Costa et al., 2025). Despite, in general manner this method reduces the influence from those processes, some

uncertainty remains, further study is need to understand vertical mixing and horizontal transport. Further details concerning the equations used to detrend the data is provided at da Costa et al. (2025). (Fig. 2).

2.8. Statistical analysis

All the datasets used here were aggregated by 0.1° grid cells, this was made using the fishnet function of ArcGis software. After matching the spatial resolution of the data, the outliers were removed using the Interquartile Range (IQR) and then the data were aggregated to a monthly time scale starting in 2009 and ending in 2019, except rainfall and fire foci. Both processes are necessary to match spatiotemporal scales and they characterize the patterns and changes in Amazon atmosphere with those remote sensing products. The rainfall and fire foci were summarized as the accumulated sum for each month in the study area.

Monthly and annual distribution graphs were constructed for XCO₂ data, XCH₄ and SIF data associated with the Amazon biome in the time series (2009–2019). Subsequently, Spearman's correlation (r) was performed with the annual means of the variables over the time series. For variables showing significant correlations with XCO₂ and XCH₄, linear regression (slope and intercept) was applied, and model accuracy was evaluated using the root mean square error (RMSE), an approach that yields analytical estimates of confidence intervals and demonstrates robustness to outliers in seasonal datasets.

Finally, the application of the Mann-Kendall test (MK) (Kendall, 1948; Mann, 1945) made it possible to verify if there is a significant trend for the data of the annual and monthly averages of XCO₂, XCH₄, SIF, and Fire foci, FRP, EVI and Rainfall in the Amazon biome, as described by Equation (5): This test was applied on the detrended data (XCO₂ and XCH₄) to understand if there is some regional driver (SIF, Fire Foci, FRP, EVI and Rainfall) increasing or decreasing its regional concentration on Amazon biome.

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(x_j - x_i) \quad (5)$$

Where x_i is n-terms with the variation ($1 \leq i \leq n$); x_j corresponds to the estimated value of n-terms. The trend test is applied to a time series x_j that is classified from $i = 1, n-1$ and x_j , which is classified $j = i + 1, \dots, n$. Each of the data points x_j is taken as a reference point that is compared to the rest of the data points x_j so that Equation (6):

$$\text{Sgn}(x_j - x_i) \begin{cases} 1 \text{ for } (x_j) - (x_i) > 0 \\ 0 \text{ for } (x_j) - (x_i) = 0 \\ -1 \text{ for } (x_j) - (x_i) < 0 \end{cases} \quad (6)$$

For the series $x = x_1, x_j, \dots, x_n$ with $n > 4$ according to the null trend hypothesis (H_0), S will demonstrate a normal distribution with variance $\text{Var}(S)$, described according to Equation (7):

Table 1

Summary of the main information for the different variables used to identify the tendency of GHGs and the fire foci, their product and their temporal and spatial resolution in this study.

Category	Variable	Product	Temporal resolution	Spatial resolution
GHGs	XCO ₂	Column-averaged dry air mole fractions of CO ₂ (Taylor et al., 2022)	3 days	10.5 km ^a
	XCH ₄	Column-averaged dry air mole fractions of CH ₄ (R. J. Parker et al., 2020)	3 days	10.5 km ^a
Vegetation	SIF	Sun-induced chlorophyll fluorescence (Frankenberg, 2022)	3 days	10.5 km ^a
	MOD13C2-Enhanced Vegetation Index	(A. Huete et al., 2002)	16 days	1 km
Climatic	Rainfall	Precipitation data from CHIRPS (Funk et al., 2015)	Daily	~5 km
	Tmmmax	Maximum temperature (Abatzoglou et al., 2018)	Daily	5 km
	WS	Wind-speed at 10m (Abatzoglou et al., 2018)	Daily	5 km
Burning vegetation	Fire foci	MYD14/MOD14 - Fire pixel location (Giglio et al., 2003)	Daily	1 km
	FRP	MYD14/MOD14 - Fire Radiative Power (Giglio et al., 2003)	Daily	1 km

^a Where: The value of “10.5 km” is the footprint size of the TANSO-FTS onboard GOSAT, and that each footprint is “~160 km” or “~280 km” apart. (Oguma et al. 2011).

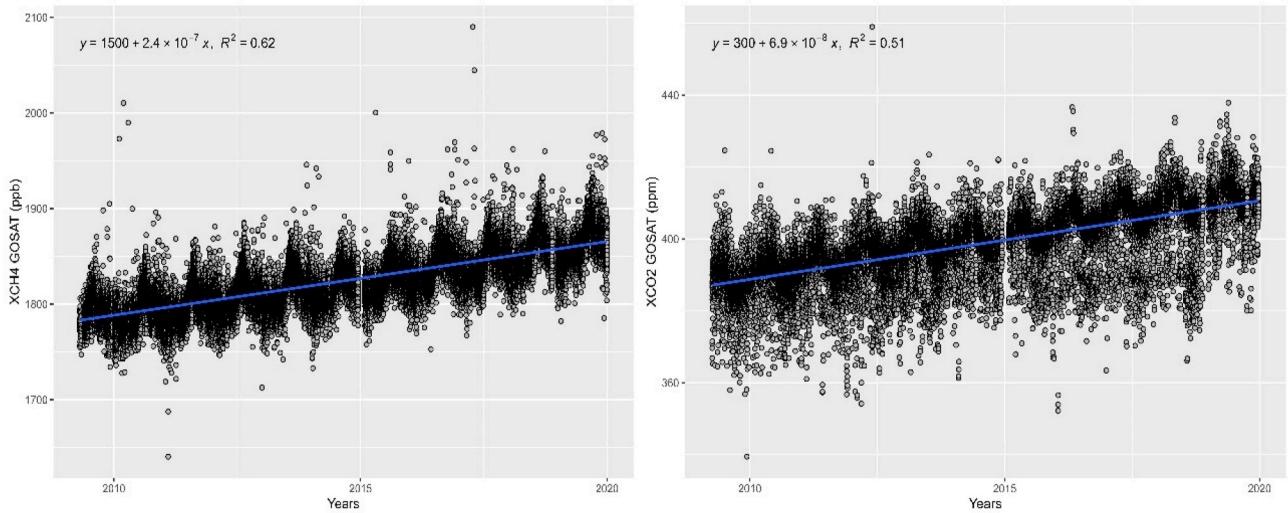


Fig. 2. Example of regression to remove the trend from data (XCO₂ and XCH₄).

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum_{i=1}^m t_i(t_i-1)(2t_i+5)}{18} \quad (7)$$

Where n is the number of observation data; t_i is the number of observation data containing identical values from the set in i ; and m is the dataset number, including equivalent values in group i from the series. The second term demonstrates a fit in the data. t_i is shown as the number of equal data up to sample i .

For the significance test S , the two-tailed test was used, in which H_0 must be rejected for higher values of Z_c (Equation (8)):

$$Z_c = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}}, & S > 0 \\ 0, & S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}}, & S < 0 \end{cases} \quad (8)$$

Then, the value of Z_c makes it possible to accept or reject H_0 , so when H_0 is accepted, there is no trend in the time series, and when there is a trend in the time series H_0 is rejected, then it accepts H_1 . Thus, this procedure establishes the hypothesis of stability of the occurrence of independent and consecutive values with the probability distribution remaining the same. The data of XCO₂, XCH₄, SIF and fire foci, FRP, EVI, and rainfall of the identified homogeneous groups were submitted to Pettitt's nonparametric test (Pettitt, 1979), which makes it possible to observe in which year the abrupt change occurred in a time series (2009–2019). The Pettitt test statistics are defined by Equations (9) and (10):

$$KT = \max |U_{t,T}| \quad (9)$$

And:

$$U_{t,T} = \sum_{i=1}^t \sum_{j=t+1}^T \text{sgn}(X_j - X_i) \quad (10)$$

Where KT is the location of the change point in the case of probability level of significance (p), which is approximately less than or equal to 5 %, described by equation (11):

$$p \approx 2 \exp\left(\frac{-6K_T^2}{T^3 + T^2}\right) \quad (11)$$

The analyses were performed using the statistical software R version 4.1.2. R is free access software, thus democratizing the access to

statistical analysis (R Development Core Team, 2021).

3. Results

3.1. Variation of greenhouse gases

The annual and monthly distributions of XCO₂ and XCH₄, for the Brazilian Amazon biome showed distinct characteristics in the analyzed time series (Fig. 3). Therefore, it is very important to note that there were outlines for some variables, even after have used the IQR technique.

XCO₂ presented annual mean values below the mean of the studied period (384.31 ± 2.05) for 2014 and 2017 (Fig. 3). For the monthly distribution, the months from May to October showed XCO₂ values above the average of the studied period (384.31 ± 2.05), and the other months were below the average (Fig. 2). It is worth noting that in 2009 and 2015, the lowest rainfall occurred, El Niño-Southern Oscillation (ENSO) (Pontes-Lopes et al., 2021) (Fig. S1); on the other hand, the highest rainfall was observed in 2011, 2013, 2014, 2018 and 2019.

The monthly distribution of XCO₂ is inversely proportional to the monthly rainfall; that is, lower values of XCO₂ were observed for the months from November to March when the highest rainfall occurred in the studied region (Fig. S1.).

The variable XCH₄ presented values above the average of the studied period (1785.18 ± 11.31) only for 2018 (Fig. 3). The monthly distribution presented values below the average for the months of the May to November (Fig. 3) drier period, when there was a reduction in monthly rainfall (Fig. S1.).

3.2. Variation SIF, Fire foci and FRP

The variable SIF recorded values below the annual average in the period studied (1.12 ± 0.16) for 2016, 2018 and 2019 (Fig. 4).

Fire foci and Fire Radiative Power data were obtained by MODerate Resolution Imaging Spectroradiometer-MODIS (Giglio et al., 2003).

For example, in 2016, there was a strong occurrence of El Niño, and a severe drought was observed in the region (Yang et al., 2018). The monthly distribution also follows the precipitation patterns, as the months from May to October record SIF values below the average (1.12 ± 0.16) (Fig. 4).

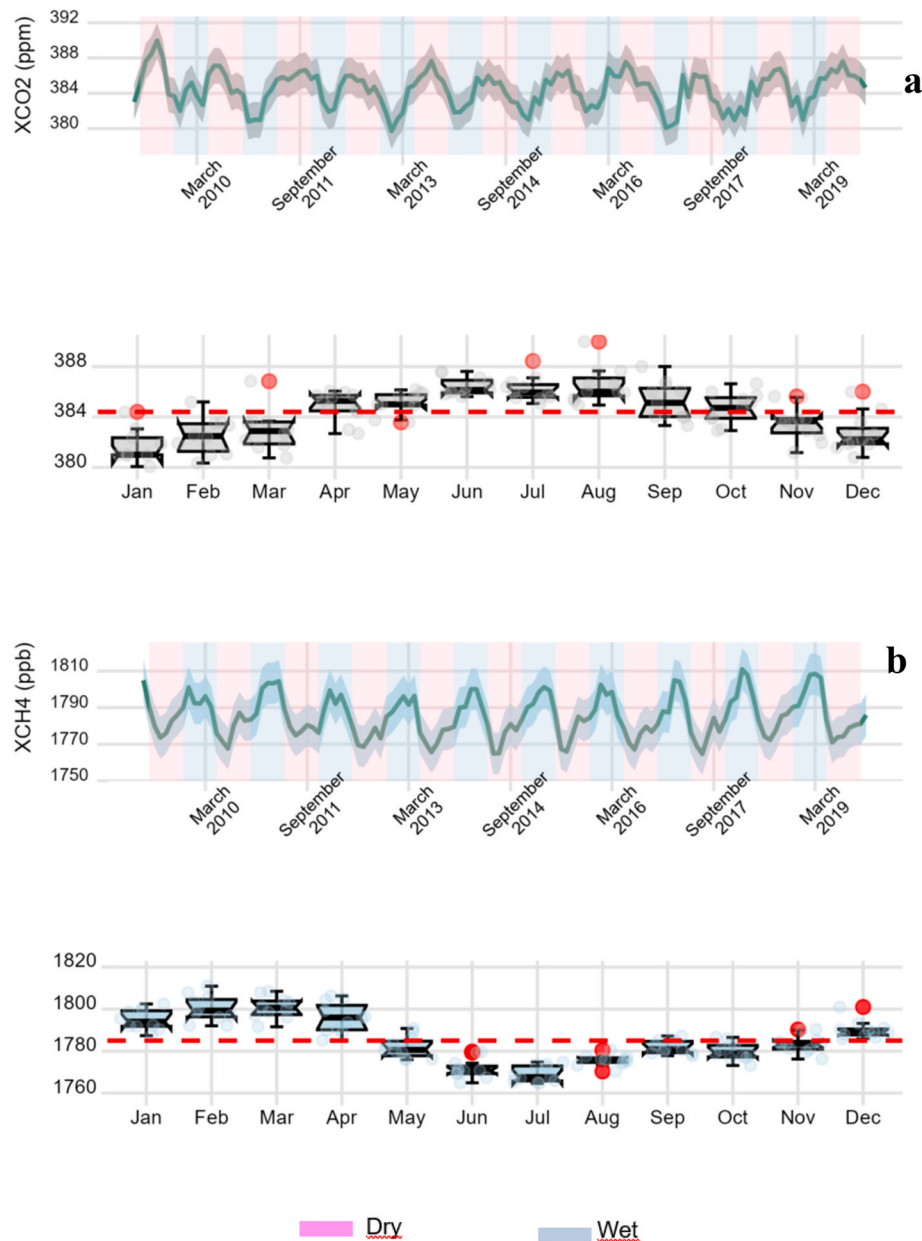


Fig. 3. Distribution of XCO₂ and XCH₄ annual and monthly without tendency for the Brazilian Amazon biome. GOSAT data were 668 (2009), 1464 (2010), 1135 (2011), 1223 (2012), 890 (2013), 726 (2014), 971 (2015), 1708 (2016), 1172 (2017), 1063 (2018) and 761 (2019). (a) Yearly and monthly average of XCO₂ data were obtained by the Japanese Space Agency - JAXA (Taylor et al., 2022). (b) Yearly and monthly average of XCH₄ GOSAT Proxy data were obtained from the University of Leicester (Parker et al., 2015). Pink shadow is dry season and blue shadow is wet season. The corresponding 95 % confidence intervals of the standard error of the mean is represented by the shadings. Dotted red horizontal line is monthly average data.

3.3. Correlation analysis of greenhouse gases with Sif, Fire foci, FRP and other meteorological parameters

Spearman's temporal correlation analysis indicates positive interactions between the variables XCO₂ with Fire Foci, FRP. On the other hand, the variables XCH₄, SIF and Rainfall established a negative linear correlation with the variables XCO₂, Fire Foci and FRP in the analysed time series (Fig. 5). Bioclimatic variables Tmax indicated a positive correlation with variables XCO₂, FRP and Fire foci. In the other hand, variables Tmax and WS indicated a negative correlation with variable XCH₄ (Fig. 5).

The XCO₂ variable is negatively related to the EVI ($r = -0.71$ and $p < 0.001$), rainfall ($r = -0.65$ and $p < 0.001$), XCH₄ ($r = -0.66$ and $p < 0.001$) and SIF ($r = -0.57$ and $p < 0.001$) variables. On the other hand, there is a significant positive correlation ($p < 0.05$) of intermediate force

between XCO₂ and FRP ($r = 0.40$ and $p < 0.001$). On the other hand, XCH₄ was strongly related to rainfall ($r = 0.8$ and $p < 0.001$); however, it showed a significant positive correlation ($p < 0.05$) of intermediate strength with the variables SIF ($r = 0.71$ and $p < 0.001$) and EVI ($r = 0.63$ and $p < 0.001$) (Fig. 4).

The analysis of standardized residuals indicated that 95 % of the residuals fell within the range of -2.5 to $+2.5$, suggesting approximate normality in their distribution. This supports the assumption that the errors were approximately normally distributed with constant variance (Fig. S2). Regression analysis of the FRP and SIF variables yielded the lowest accuracy values by RMSE (1.85 and 1.70, respectively) (Fig. 6).

The Mann-Kendall test was not significant at the 5 % probability level ($p < 0.05$), statistically corroborating the confirmation of the absence of increasing or decreasing monotonic behavior in the annual average for the variables XCO₂, XCH₄, SIF, EVI, rainfall, FRP and fire

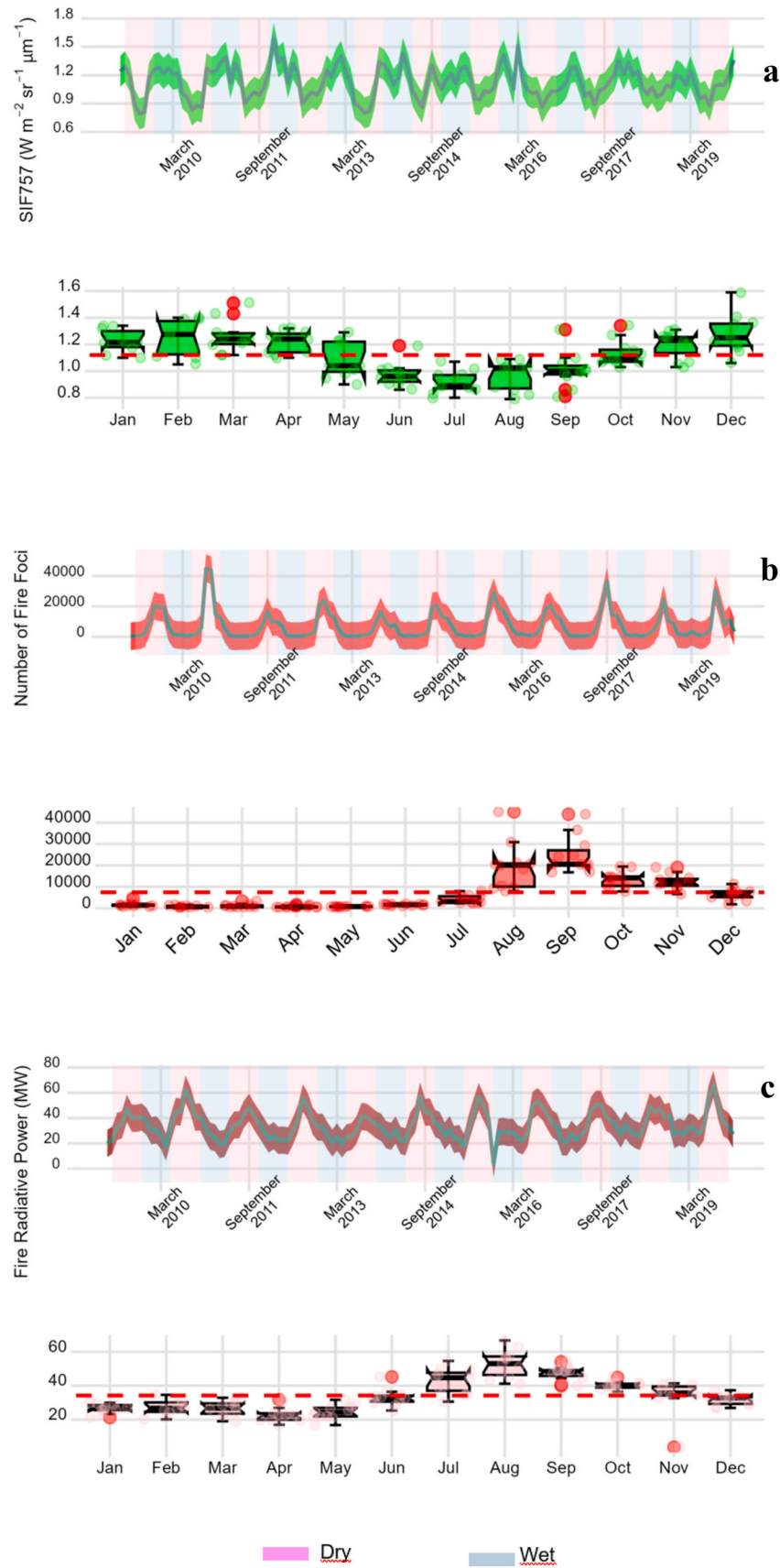


Fig. 4. Distribution of Count Fire foci, Fire Radiative Power (FRP) and SIF GOSAT annual and monthly without tendency for the Brazilian Amazon biome. (a) Monthly and annual average Sun-Induced Chlorophyll Fluorescence Data (SIF 757) were obtained by the GOSAT satellite (Frankenberg, 2022). (b) Monthly and annual average of Number.

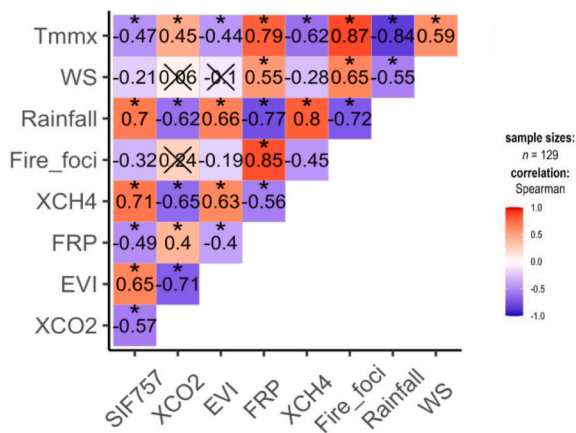


Fig. 5. Correlation of the monthly mean variables XCO₂, XCH₄, SIF, Rainfall, EVI, FRP, Fire foci, Tmmx (Tmax) and WS evaluated between 2009 and 2019 for the Brazilian Amazon biome. The symbol * represents $p < 0.001$ and symbol X represents significant not.

foci analyzed in the time series (Table S3). Therefore, it was not necessary to perform the Pettitt test, justified by the absence of a trend in the annual averages in the time series.

For trend analysis of monthly averages, XCO₂ for the Mann-Kendall test was significant at the 5 % probability level ($p < 0.001$), indicating the presence of a trend for April along the time series (Table S4), and the application of the Pettitt test to the annual time series of fire counts revealed a statistically significant breakpoint in the year 2014 ($p < 0.01$). This indicates an abrupt shift in the fire regime within the Amazon biome starting from that year (Fig. S3).

For the other months, this trend was not observed. Thus, through a regression analysis, the averages April there were an increase (confirmed by Pettitt's test) between the analyzed time serial, when comparing the monthly averages of time series (11 years), there was an increase after 2013 an increase above 1.3 ppm (XCO₂) to averages of April identified as a change point (Pettitt test) (Fig. S3). In the same way, the variable FRP had an increase for 12.4 MW (April) and 23.9 MW (July) (Fig. S4). Our results were in agreement with inventory emissions VFEI, noted an increase of emissions of CO₂ in month of April in the period from 2012 to 2019 (Fig. S5).

It is also worth noting that for the trend of monthly averages, the variable FRP for the Mann-Kendall test was significant at the 5 % probability level ($p < 0.001$), indicating the presence of a trend for the month of April along the time series. In the same way as for the month of July (Table S4), the FRP variable was significant at the 5 % probability level ($p < 0.001$) (Fig. S4). Thus, the Pettitt test was significant at the

level of 5 % probability for the month of April ($p < 0.005$) and for the month of July ($p < 0.005$), indicating continuous growth after 2014 (Fig. S4 and Table S5). For the other months, this trend was not observed. Our results were validated with inventory emissions VFEI, there was an increase of FRP after month of April and July in the period de 2012 to 2019 (Fig. S5).

4. Discussion

The time series analysis from 2009 to 2019 revealed key interactions between atmospheric carbon concentrations, precipitation, photosynthetic activity, and biomass burning in the Amazon biome. Notably, the highest atmospheric CO₂ concentrations (XCO₂) coincided with periods of reduced precipitation and diminished photosynthetic activity during the dry season (Fig. 3a; Fig. S1). This suggests that the combined effect of water stress and reduced photosynthesis may be driving the forest toward a loss of its carbon sequestration capacity, justified by the lower availability of water and lower photosynthetic rate of the vegetation. This seasonal pattern is in agreement with the carbon flux seasonality reported by Gatti et al. (2023); Basso et al. (2023).

Our results indicate that reductions in solar-induced chlorophyll fluorescence (SIF) and the Enhanced Vegetation Index (EVI) are positively correlated with increased XCO₂. During the dry season, lower SIF values reflect decreased photosynthetic activity, likely due to water limitations (Albright et al., 2022; Wu et al., 2017). A significant negative temporal correlation between XCO₂ and rainfall ($r = -0.62$, $p < 0.001$; Fig. 5) further supports this relationship. Similarly, XCO₂ showed negative correlations with EVI ($r = -0.71$, $p < 0.001$) and SIF ($r = -0.57$, $p < 0.001$), underscoring the interplay between vegetation dynamics, photosynthetic activity, and atmospheric CO₂ levels.

Consequently, the dry season exhibits a higher occurrence of fire foci and higher FRP values, which are positively correlated with increased XCO₂. This indicates that biomass burning emissions contribute to the higher XCO₂ observed during this period, in addition to the reduction in photosynthetic uptake (Fig. 6). The most severe droughts were caused by the El Niño-Southern Oscillation (ENSO) and the Atlantic Multi-decadal Oscillation (AMO) in the Amazon region (Marengo et al., 2018). The reduction of precipitation in the Amazon (Gatti et al., 2014), the intensification of land use change (Silva et al., 2020), and the flammability of the Forest (Nepstad et al., 2004) are impacted by these more extreme weather events affecting the regional carbon cycle (Smith et al., 2014).

In contrast to XCO₂, XCH₄ showed no positive correlation with fire radiative power (FRP) or fire counts. Instead, XCH₄ was strongly correlated with rainfall ($r = 0.80$, $p < 0.005$), identifying soil moisture as the primary control on Amazonian CH₄ emissions. This is consistent with findings that flooded areas dominate (>70 %) the regional CH₄ budget, emitting 33.9 ± 10.8 Tg CH₄ yr⁻¹ (Basso et al., 2021), with

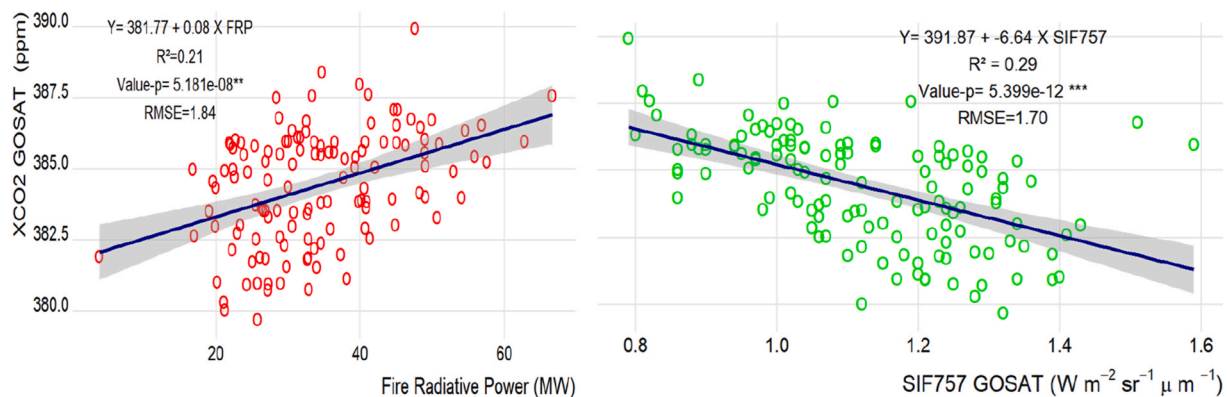


Fig. 6. Linear regression between Fire Radiative Power (MW), Sun-induced Fluorescence -SIF GOSAT ($\text{W m}^{-2} \text{sr}^{-1} \mu\text{m}^{-1}$) and XCO₂ GOSAT (ppm) variables. The corresponding 95 % confidence intervals of the standard error of the mean is represented by the shadings.

emissions amplified during wetter periods such as La Niña events (Nisbet et al., 2014). The persistence of elevated dry-season concentrations implies a secondary source from biomass burning. Precise quantification of individual sources (wetlands, fires, livestock) necessitates atmospheric inversion modeling, a challenge complicated by CH₄'s long atmospheric lifetime (~9 years), which integrates local emissions with long-range transport.

While our analysis focuses on the immediate atmospheric response to fire emissions, it is imperative to consider the long-term carbon cycle implications. The repeated burning and degradation of forests not only release carbon immediately but also damage the ecosystem's future capacity to sequester carbon, effectively weakening the Amazon's role as a carbon sink over time. This study captures the acute pulse of emissions, but the chronic degradation of the sink is an equally vital consequence of the fire crisis in the Amazon.

Globally, CH₄ emissions from wetlands represent 65 % of total emissions and are influenced by factors such as temperature, carbon availability, and water table depth (Saunois et al., 2016; Bloom et al., 2012). Similar patterns were observed in East Africa, where rainfall anomalies impacted wetland CH₄ emissions by altering water table levels (Lunt et al., 2021). These dynamics further highlight the significant role of the rainy season and wetland areas as primary sources of CH₄ in the Amazon. Similarly, rainfall anomalies contribute to altering the water table, directly affecting CH₄ emissions in the wetlands (Pandey et al., 2017).

The relationship between XCO₂ and SIF, analyzed through linear regression, revealed a slope of $-6.64 \text{ ppm (W m}^{-2} \text{ sr}^{-1} \text{ } \mu\text{m}^{-1})$, which is comparable to but slightly higher than values reported in shorter time series (Costa et al., 2021). The coefficient of determination ($R^2 = 0.29$) suggests a moderate relationship between these variables, with precipitation strongly influencing SIF, and CO₂ concentrations reflecting the balance of photosynthesis and respiration (Albright et al., 2022). These findings are consistent with observations during the 2015–2016 El Niño event, which caused precipitation reductions and suppressed photosynthesis (Liu et al., 2017; Yang et al., 2018; Koren et al., 2018).

From 2009 to 2019, our analysis shows that FRP was positively correlated with XCO₂, reinforcing the role of biomass burning in elevating XCO₂. Fire activity is particularly impactful during the dry season (May–October), contributing significantly to greenhouse gas emissions (Silva Junior et al., 2019). The high correlation between FRP and fuel consumption rates underscores its utility as an indicator of gas emissions, including CO₂ (Freeborn et al., 2008; Wooster et al., 2005).

XCO₂ may interact with vegetation and soil respiration, therefore the same interaction is not observed for the FRP variable, possibly justifying greater linearity in the time series (2009–2019) of the FRP variable. On the other hand, a monotonic increase in the variables XCO₂ and FRP was observed in the April averages, while FRP alone exhibited an increase in July. Notably, a linear relationship between changes in FRP and variations in XCO₂ was not identified during the peak of the dry season (July/August). This lack of linearity may be attributed to consistently high GHG concentrations during this period, which exhibit minimal variation. However, a more concerning possibility is the earlier onset of fires. Our results suggest a trend that could be explained by an extended dry season in the southern region and a reduced rainy season in the north, particularly in April—even though no significant reduction in April rainfall was observed in the time series. This earlier fire activity, potentially driven by anthropogenic ignition rather than climate alone, could accelerate vegetation changes from forest to savanna (Bush, 2017; Nobre et al., 2016).

Our study demonstrated an increase in XCO₂, revealing a significant correlation between atmospheric CO₂ concentrations and fires in the Brazilian Amazon biome. The findings are supported by the VFEI inventory ($\text{kg m}^{-2} \text{ s}^{-1}$), which indicates an increase in gas emissions in the region during the analyzed period. The Amazon biome, recognized as the largest tropical forest on Earth, has demonstrated significant relevance as a critical carbon sink over recent decades, playing a pivotal role

in global carbon cycling and climate regulation. (Maia et al., 2020). However, this carbon sink could possibly be in decline by the increase of land use change, fires and droughts in the biome (Gatti et al., 2021). An important uncertainty in our analysis is the influence of atmospheric transport, which was not explicitly accounted for here. Although air circulation in the Amazon region is generally homogeneous with air masses flowing from the Atlantic coast to the west direction. Our analysis provides insights about changes in atmospheric levels possibly caused by fire occurrence. A complementary analysis including these data in an atmospheric inverse modelling would be beneficial to confirm our findings.

Based on our results, fires increase the concentrations of GHGs (CO₂) and reverse the role of the Amazon Forest from a sink to a source of CO₂ to the atmosphere. In addition, we mentioned that our findings are in agreement with previous publications that report carbon emissions in this region and relate with biomass burning, with use of carbon fluxes from CO₂ vertical profile observations and estimates fluxes using Mass balance technique and atmospheric inverse modelling (Basso et al., 2023; Gatti et al., 2021).

The identification of 2014 as a change point is not merely a statistical artifact but likely reflects a confluence of critical climatic and anthropogenic drivers. The year 2014 marked the onset of the strong 2014–2016 El Niño episode, which led to pronounced drought conditions in the northern Amazon, increasing vegetation flammability and fire risk (Jiménez-Muñoz et al., 2016). Concurrently, this period in Brazil was characterized by a notable relaxation in environmental law enforcement and a surge in deforestation rates, creating conditions ripe for human-induced fires (INPE, 2022; Escobar, 2019). The synergy between extreme drought and increased anthropogenic ignition sources likely compounded fire activity, resulting in the detected regime shift.

Finally, from a climate point of view, as it shows that the Amazon biome may be very close to the tipping point (Lovejoy & Nobre, 2019), in addition to directing discussions on the processes of savannization of the biome (Nobre et al., 2016), mainly the southern and eastern portion of the Amazon biome. Thus, our results highlight a possible anthropogenic signal in the earlier onset of fires, particularly in April, which is not explained by rainfall anomalies. This pattern coincides with documented increases in land-clearing activities and reduced enforcement of environmental regulations in recent years (Gatti et al., 2023).

The earlier rise in FRP and XCO₂ could reflect an adaptive strategy by land users to burn before stricter fire bans in the core dry season, or to take advantage of transitional weather windows when vegetation flammability is high but conditions are less extreme than in August–September. This shift, if confirmed, could have significant implications for carbon emission estimates, as early fires may differ in combustion completeness and atmospheric transport patterns.

5. Conclusion

This study provides the first satellite-based, multi-sensor evidence of a shift toward earlier fire activity in the Brazilian Amazon, with significant increases in FRP and XCO₂ detected in April (both variables) and July (FRP only) between 2009 and 2019. Despite our study did not distinguish natural and non-natural fire-activity, no changes in general rainfall pattern for the whole Amazon was observed, which suggest that this early increase in fire activity, especially in April since is far from the natural fire incidence peak (July–October), is linked to anthropogenic activity. However, further studies have to better elucidate the nexus between land use change – fire outbreaks – early atmospheric GHG concentration increase. In contrast, XCH₄ dynamics were primarily linked to hydrological variability, with strong positive correlations with rainfall, highlighting wetland emissions as the dominant source.

Despite our study is limited to a biome general assessment, without considering different fire origins, atmospheric transport modes (vertical and horizontal) and a global fire emission factor, our results highlight the potential weakening of the Amazon's role as a global carbon sink and

its increasing vulnerability to climate–land-use feedbacks. Future work should integrate inverse atmospheric modeling, higher-resolution fire emission inventories, and socio-environmental datasets to confirm the drivers of early fire occurrence and assess their long-term implications for biome resilience and global carbon budgets.

From a policy perspective, the observed advancement of fire activity into the pre-dry season has significant implications for enforcement strategies and emission reduction targets. If confirmed as a persistent trend, this shift could alter combustion completeness, emission factors, and atmospheric transport patterns, ultimately affecting the accuracy of carbon budget estimates. Early detection of such shifts can inform adaptive fire management, enhance climate change mitigation strategies, and contribute to achieving national and international climate commitments. Here we demonstrated, from a general pattern perspective, that the integration of GOSAT, MODIS, VIIRS, and vegetation indices demonstrates a scalable framework for detecting and monitoring such shifts, with potential applications in fire early-warning systems and emission mitigation efforts. Further efforts have to be made to understand the feasibility in better spatiotemporal resolutions.

CRedit authorship contribution statement

Luciano de Souza Maria: Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Luis Miguel da Costa:** Validation, Software, Investigation, Data curation. **Marcelo Odorizzi Campos:** Resources, Methodology, Formal analysis, Conceptualization. **Alan Rodrigo Panosso:** Visualization, Validation, Software, Project administration, Investigation. **Luana Santamaria Basso:** Writing – original draft, Validation, Resources, Formal analysis, Conceptualization. **Carlos Antonio da Silva Junior:** Writing – review & editing, Visualization, Resources, Investigation, Data curation. **Newton La Scala:** Writing – review & editing, Validation, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jastp.2025.106652>.

Data availability

Data will be made available on request.

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